

version control system including Bitbucket, GitHub, GitLab, SourceForge, and more. For source control management (SCM) purposes, we use GitHub.

## Continuous integration

Continuous integration is a phase in SDLC where developers' code is integrated to create an artefact. Each integration of code undergoes the building process, automated integration testing, and quality assurance. Once the code is successfully built, the artefact can be deployed to remote systems using the automated deployment tools integrated with the continuous integration tools.

Jenkins is one such open source DevOps tool used for continuous integration. It automates various parts of the SDLC, including fetching the developer's code from GitHub, testing that code, building the code in an artefact, and further calling automation deployment tools. Jenkins can integrate most of the tools while facilitating continuous testing, building, and integration. The workflow of Jenkins is depicted in Figure 2.

## Continuous testing

Testing plays a crucial role in the entire SDLC to ensure what is built meets the real world requirements. Testing helps reduce the risk of failures in production and provides immediate feedback during assessment. It involves validating APIs, testing for security vulnerabilities, and assessing the performance of the application under stress.

## Continuous deployment

Continuous deployment involves deploying code to remote servers without complications. Traditionally, deployment can be a complex and time-consuming task; it must be reliable in order to avoid project service failures

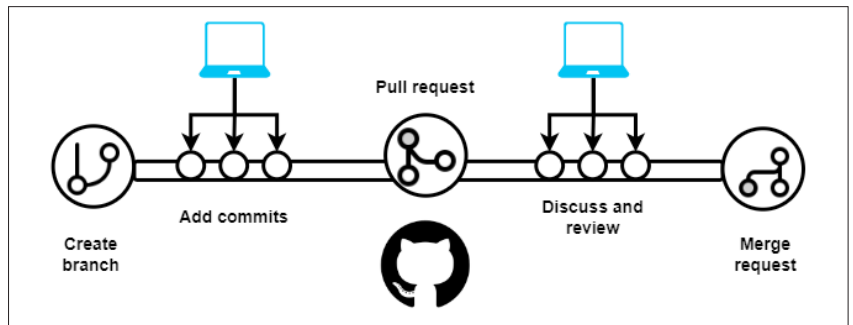


Figure 1: GitHub workflow

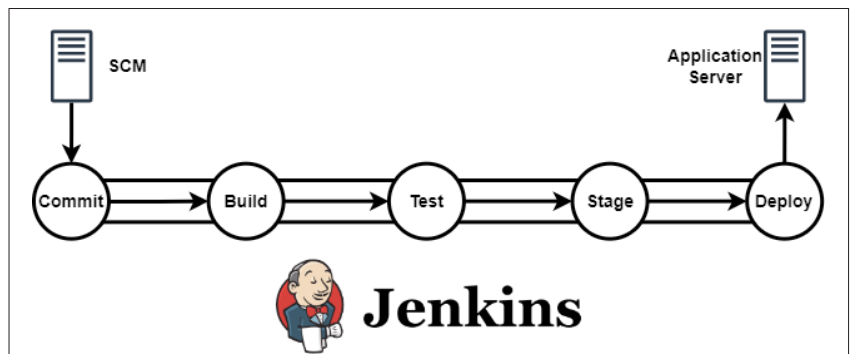


Figure 2: Jenkins workflow

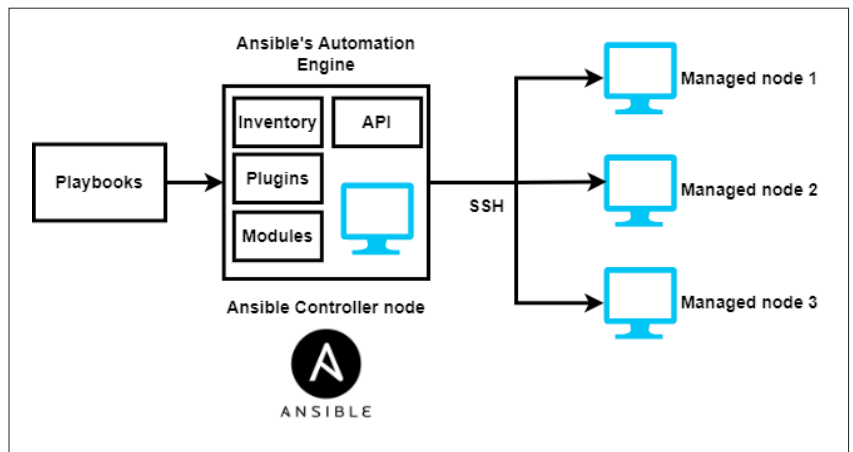


Figure 3: Ansible workflow

or server downtime. To achieve seamless continuous delivery and immediate autonomous deployment to the production environment, we can utilise Ansible, a powerful DevOps tool.

Ansible is open source software that provides configuration management, software provisioning, and deployment capabilities. With Ansible, code deployment, configuration and

maintenance become easier tasks. The workflow of Ansible is illustrated in Figure 3.

## Continuous feedback

Continuous feedback involves capturing the user experiences and insights of the application to strategise future actions and improvements. It's the continuous interaction of application users with